

Question

A box contains 10 red cards and 10 blue cards, each card being numbered with $1, 3, 3^2, \dots, 3^9$ once. The total sum of the numbers on both colored cards is denoted as $$$$$. For a given positive integer n , if it is possible to select a number of cards from the box such that the sum of their numbers is exactly n , it is called a " n selection scheme." The number of different n schemes is denoted as $f(n)$. Find the value of $P = \sum_{n=1}^{1000} f(n)$.

Qwen3-8B [full]

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